Lebanese Pizzeria Database Management System

A comprehensive database solution designed to streamline inventory, menu, and sales management for Lebanese pizzeria operations

Final Project Presentation | METCS:779 Advanced Database Management

Agenda

This presentation covers our complete database solution for Lebanese pizzerias, from concept to implementation.

01

Business Use Case

Overview of the Lebanese pizzeria business and data management challenges

02

Database Design

Core entities, relationships, and data integrity implementation

03

Transaction Management

Stored procedures, triggers, and automated inventory systems

04

Query Optimization & Cloud Deployment

Performance tuning, reporting solutions, and cloud architecture

05

Challenges & Future Work

Lessons learned and roadmap for system evolution

Business Use Case & Challenges

Lebanese pizzerias face unique operational challenges that require specialized database solutions:

High-Volume Ingredient Tracking

Tracking multiple specialized ingredients with varied shelf lives, from za'atar and sumac to fresh vegetables and imported cheeses

Complex Menu Relationships

Managing traditional items like manakish, lahm bi ajeen, and modern fusion pizzas with customization options

Real-Time Data Requirements

Need for instant inventory updates during peak hours when hundreds of transactions occur simultaneously

Database Design Overview

Our relational database architecture focuses on five core entities with carefully designed relationships to ensure data integrity and operational efficiency.

The schema implements **third normal form (3NF)** to eliminate redundancy while maintaining query performance. Primary and foreign keys enforce referential integrity, with particular attention to the many-to-many relationship between sales and menu items through the SaleDetails junction table.

Schema Normalization & Data Integrity

Normalization Strategy

- Inventory items separated from suppliers (2NF)
- Menu prices extracted from transactional data (3NF)
- Customer information isolated from sales records
- Ingredient quantities normalized across recipes

This approach eliminated data anomalies while preserving the ability to reconstruct historical pricing information.

Integrity Constraints

```
CREATE TABLE inventory_transactions (
    transaction_id INT AUTO_INCREMENT PRIMARY KEY,
    inventory_id INT,
    transaction_type ENUM('Sale', 'Purchase', 'Adjustment', 'Waste')
NOT NULL,
    quantity_change DECIMAL(10,2),
    transaction_date DATETIME NOT NULL,
    reference_id INT,
    notes TEXT,
    FOREIGN KEY (inventory_id) REFERENCES inventory(inventory_id)
);
```

Transaction Management & Automation

Our system maintains transactional consistency through a series of automated processes:

Sale Initiated

Customer order entered into POS system, triggering a new transaction in the Sales table

Inventory Updated

Triggers automatically decrement inventory quantities based on menu item recipes

Alerts Generated

Low stock levels trigger SMS notifications to managers via Twilio API integration

Query Optimization & Reporting

We developed a suite of optimized views and stored procedures that deliver critical business intelligence with minimal performance impact.

Daily Sales Summary

Aggregated view with indexing on date fields, delivering sub-second response times even with **potentially** 10,000+ daily transactions (If the business decides to scale up)

Inventory Status Report

Real-time view with projected depletion dates based on 30-day consumption trends

Menu Performance Analysis

Comparative sales metrics with profit margin calculations per menu item

```
CREATE OR REPLACE VIEW v_top_selling_items AS
SELECT
    m.menu_id,
    m.item_name,
    SUM(sd.quantity) AS total_sold,
    SUM(sd.quantity * sd.price_each) AS total_revenue
FROM sale_details sd
JOIN menu m ON sd.menu_id = m.menu_id
GROUP BY m.menu_id, m.item_name
ORDER BY total_sold DESC;
```

Cloud Deployment & Scalability

- ## Current Cloud Database Implementation

  Deployed on freesqldatabase.com. With further support from Ajeen, can be deployed on AWS

- ## Multi-Region Availability

  Worldwide access for business continuity and fulfills business requirement to be accessed by owner in Lebanon

- ## API Integration Layer

  RESTful API services connect inventory status to Twilio for SMS notifications and enable future integration with delivery platforms

Key Learnings & Skills Acquired

This project provided comprehensive hands-on experience, reinforcing key database concepts explicitly covered in Modules 0.1, 1, 2, 3, and 4:

- SQL Data Definition Language (DDL) and Data Manipulation Language (DML)
- Database Normalization Principles (up to Third Normal Form)
- Utilizing Views for Business Logic and Reporting
- Effective Primary and Foreign Key Design
- Cloud Database Deployment & Connectivity
- Referential Integrity Enforcement

Challenges & Lessons Learned

Foreign Key Constraint Violations

Initial data migration failed due to circular dependencies. Solution: Implemented a staged migration process with temporary constraint disabling and validation steps.

Compatibility Issues with FreeSQLDatabase.com

The free SQL database (freesqldatabase.com) had limitations with certain advanced SQL features and past version syntax. Solution: Adapted queries and triggers to compatible syntax and offloaded complex logic to the application layer.

Anticipating Query Performance at Scale

While current data volume is limited, future reports may slow significantly with data accumulation. Solution: Plans include implementing table partitioning by date and creating aggregate tables for historical data to ensure long-term performance.

Conclusion & Future Work

Potential Benefits

- Enable significant reduction in inventory waste through accurate tracking and forecasting
- Potential to decrease order processing time, streamlining operations
- Facilitate data-driven menu optimization, potentially increasing average ticket size
- Offer the capability to automate a majority of manual inventory management tasks

Roadmap for Evolution

1. **Q2 2023** — Implement real-time dashboard with Power BI integration
2. **Q3 2023** — Develop customer loyalty program database extension
3. **Q4 2023** — Integrate with third-party delivery platforms via API
4. **Q1 2024** — Implement predictive analytics for demand forecasting

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